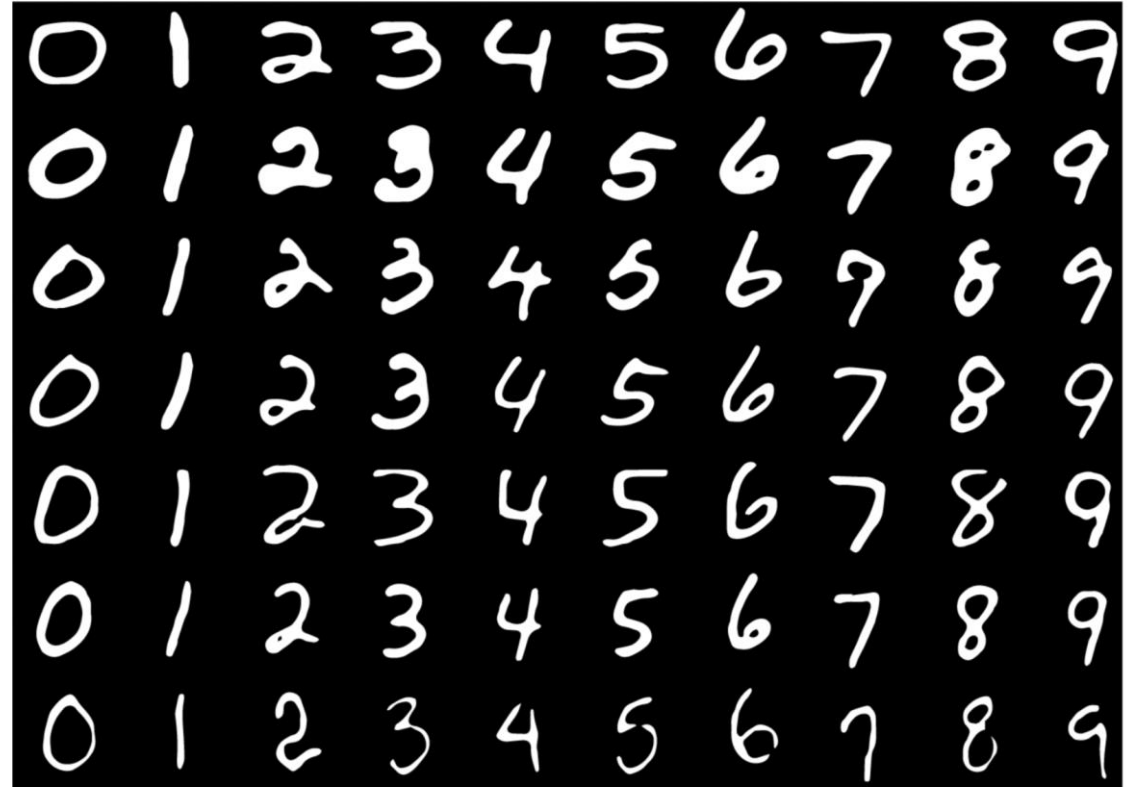


Lecture 13

Open Datasets

MNIST

- Handwritten digits dataset, 0 to 9 classes
- Training images: 60,000
- Test images: 10,000
- Gray-scale images (single channel)
- Image size: 28x28



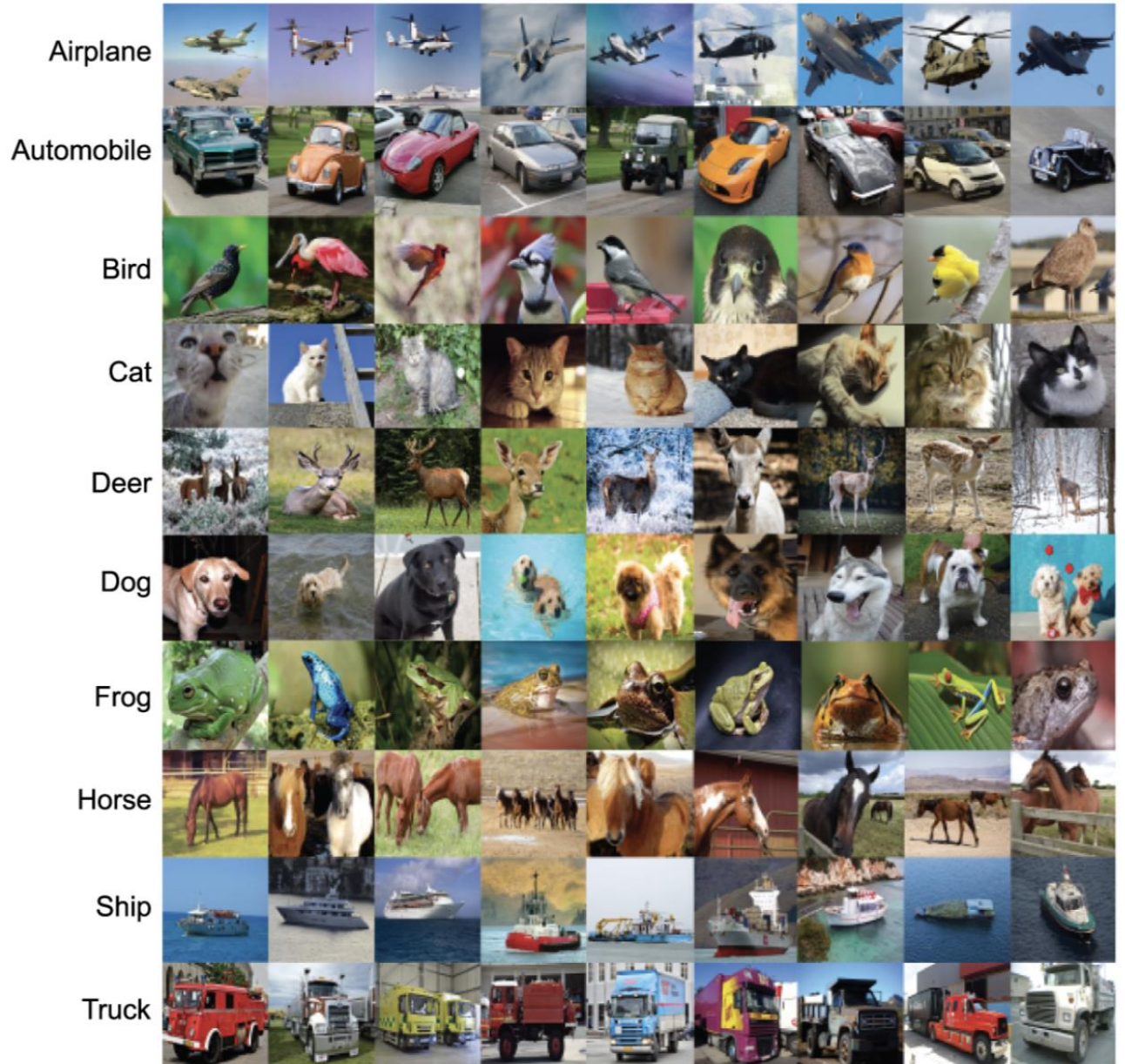
Fashion MNIST

- 10 Fashion clothing classes: t-shirt/top, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, and ankle boot
- Training images: 60,000
- Test images: 10,000
- Gray-scale images (single channel)



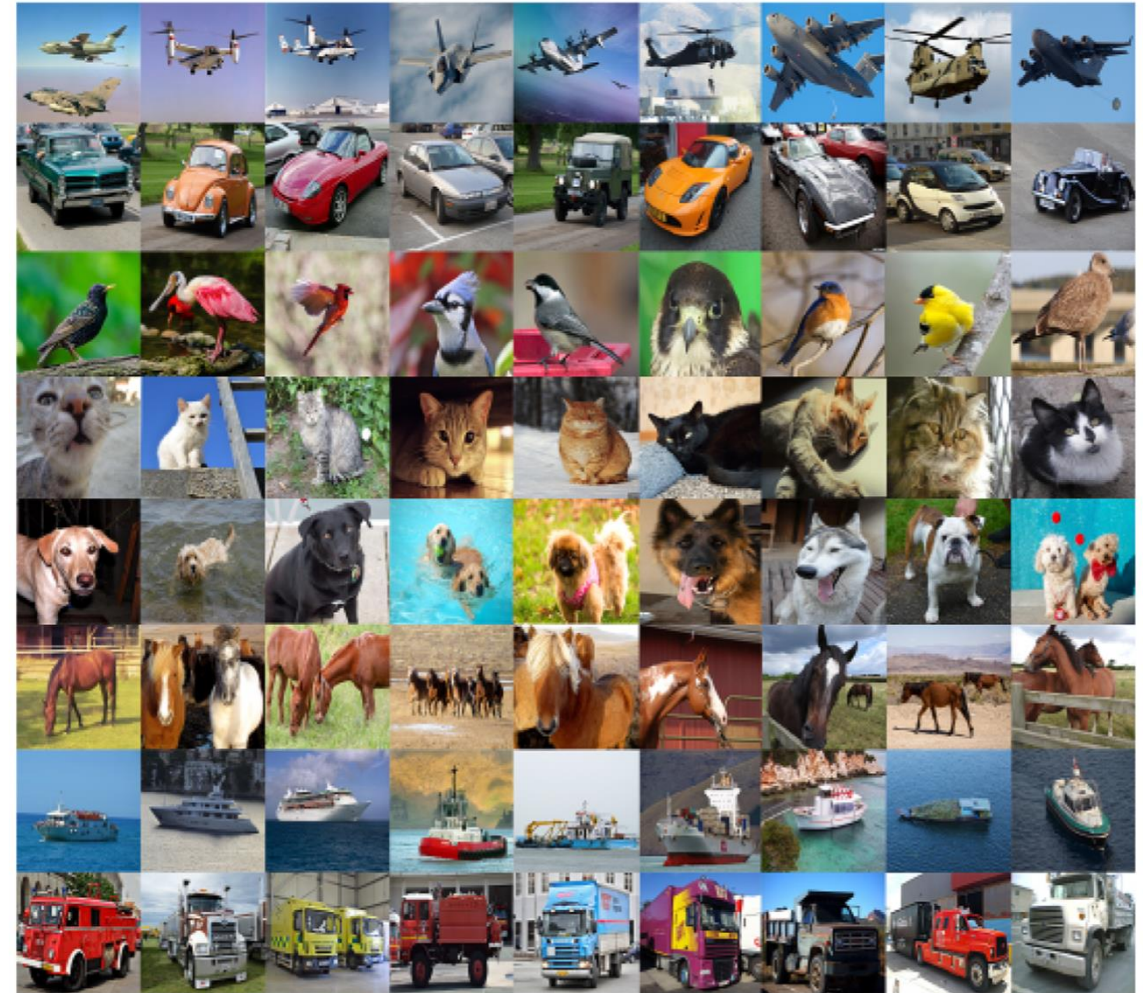
CIFAR

- CIFAR-10
 - Not perfectly centered as MNIST datasets
 - Contains on 10 classes
 - Training images: 50,000
 - Test images: 10,000
 - Color images
 - Image size: 32x32
- CIFAR-100
 - Contains on 100 classes with 600 images per class
 - 100 classes grouped into 20 super classes
 - Each image has 2 labels
 - Fine label – for the class
 - Coarse label – for the superclass

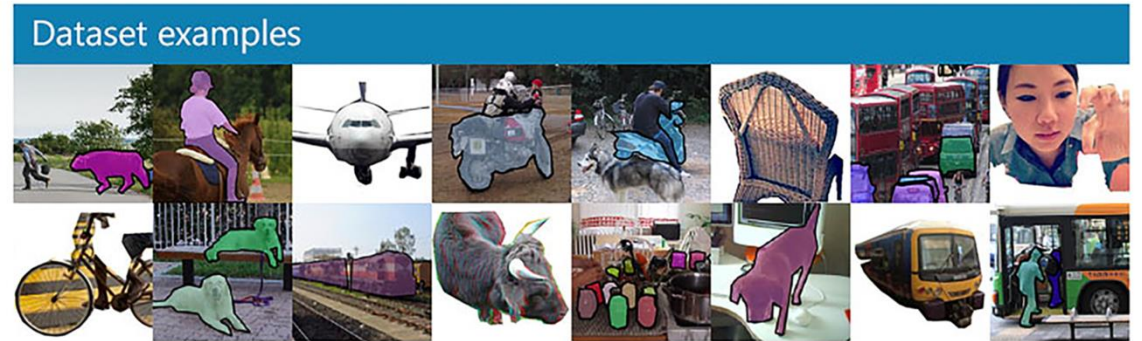


ImageNet

- Considered as benchmark for CV model evaluations
- Classification, object detection
- Contains 22000 categories
- Images collected from the web and labeled by humans
- Number of images: more than 14 million
- The images are organized by WordNet hierarchy
 - each meaningful word/phrase in WordNet is called a synonym set (synset for short)
 - 1,000+ images per synset
 - E.g. animal → mammal → dog → golden retriever
- usually refers to the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) when talking about ImageNet

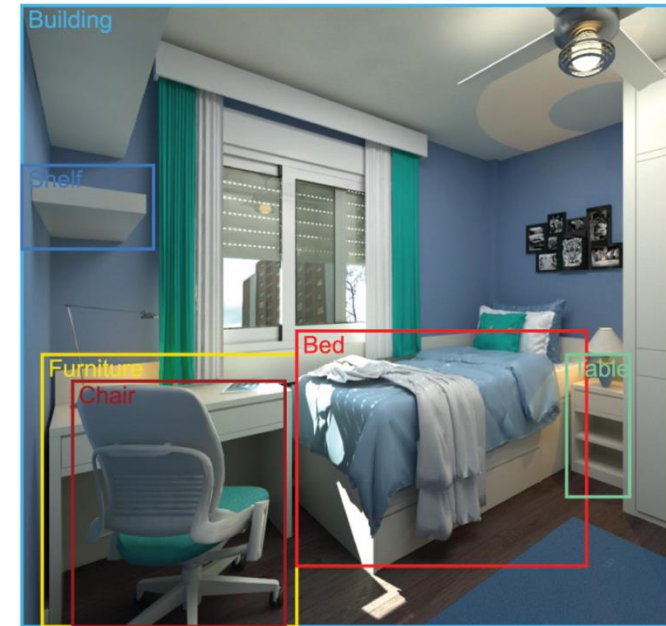
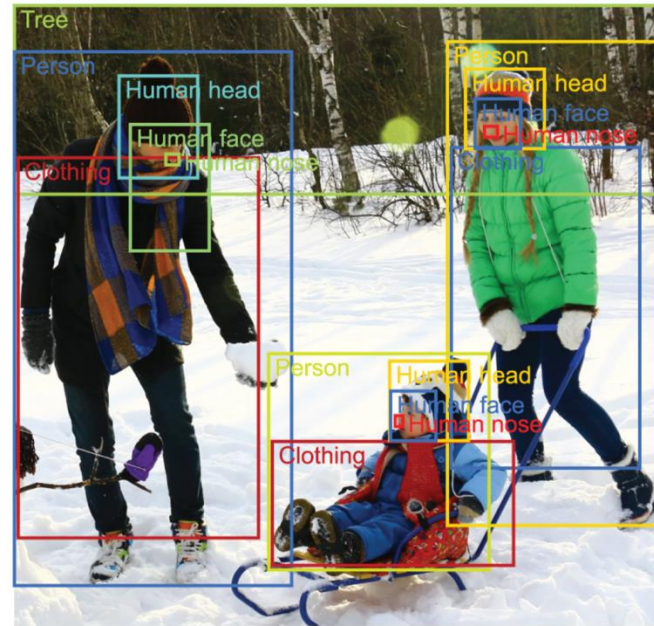


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Google Open Images

- contains more than 9 million images
- object detection, image classification, instance segmentation
- Includes images with complex scenes that span thousands of classes of objects
- more than 2 million of these images are hand-annotated with bounding boxes
- Open Images has a challenge called the Open Images Challenge



Kaggle

- Contains a number of datasets
- Hosts various ML and DL challenges

